UM-3 analysis

The x0 mass mechanism, dirac16complex as dark matter and as dark energy, and what the author may have left out

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Analysis of UM-3, 2026-10-24

Repository once-ere/Dirac_claude, folder universe_model_3

Contents

1 How to read this document **1**
1.1 What this document contains . . . 1
1.2 The status labels . . . 2
1.3 Every equation is followed by its symbols . . . 2
1.4 The answers in brief . . . 2

2 The x_0 mass mechanism **3**
2.1 The period of x_0 and the proper length of the hidden direction . . . 3
2.2 What the hidden direction produces instead . . . 4
2.3 The answer . . . 6

3 dirac16complex as dark matter **7**
3.1 Stability . . . 7
3.2 Gravitation on the fold . . . 9
3.3 The equation of state of the Kohn-Sham states . . . 9
3.4 The relic density, direct detection and the window . . . 11
3.5 The answer . . . 14

4 dirac16complex and the dark energy **14**
4.1 The answer . . . 16

5 What the author may have left out **16**

1 How to read this document

1.1 What this document contains

The author asks for three analyses of the universe model UM-3: the mass mechanism along the hidden coordinate x_0 (the period $\pi/(6H)$ of the metric and the uncertainty relation), whether dirac16complex can be the dark matter, and whether it can be the dark energy. He also asks to be told what he may have left out. This document gives the four answers. It is self-contained: every equation it uses is written in it (with the tag it carries in every UM-3 document, FE-n for a field equation, or A-n for an equation of this document), and every symbol of every display equation is defined right after it. The sections are:

- **The answers in brief.**
- **The x_0 mass mechanism**: the period of x_0, the proper length of the hidden direction, the spectra that replace the tower of a compact dimension, and the uncertainty relation.
- **dirac16complex as dark matter**: stability, gravitation on the fold, the equation of state of the Kohn-Sham states, the relic density, direct detection, and the window of parameters that survives.

- **dirac16complex and the dark energy**: why the gas of bound states cannot be dark energy, what the dark energy of UM-3 is, and how it compares with the baryon acoustic oscillations measured by DESI DR2 (the second data release of the Dark Energy Spectroscopic Instrument).
- **What the author may have left out**: the additions that UM-3 needed, and the open problems.

1.2 The status labels

Every result carries one of the labels **AUTHOR** (from the author's notebooks `4+4.nb` and `Dirac_Einstein-Lovelock_4+4_Nash.nb` and his order of 2026-10-24), **RECORD** (from the record of this repository: the Lagrangian of dirac16complex with its canonical spin connection and contact self-interaction, the Kohn-Sham treatment), **STANDARD** (textbook physics or a published measurement, named), **ASSUMED** (a choice of UM-3 without derivation), **INVENTED** (a new interaction of UM-3: the portals of dirac16complex), **DERIVED** (computed and verified in UM-3, with the verification stated), **ESTIMATE** (an order-of-magnitude evaluation) and **OPEN** (not settled).

1.3 Every equation is followed by its symbols

Every display equation is followed by the list of EVERY symbol that appears in it, with its meaning, unit (natural units $\hbar = c = 1$), value and status. The author's coordinates are $X = \{x_0, x_1, \ldots, x_7\}$: x_0 the hidden space-like coordinate (the metric depends on it through $z = 6Hx_0$), x_1, x_2, x_3 ordinary space, x_4 the time, x_5, x_6, x_7 three extra time-like coordinates; the signature is $(+,+,+,+,-,-,-,-)$. H is the author's constant (an inverse length), a free parameter of UM-3. The Standard Model lives on the fold of the metric (the surface $z = \pi/2$, where the proper coordinate σ of the hidden direction is zero), dirac16complex and gravity in the bulk.

1.4 The answers in brief

1. **The x_0 mass mechanism.** The coordinate x_0 has the period $\pi/(6H)$, but one period is a hidden direction of infinite proper length, warped towards the fold: there is no compact dimension, no Kaluza-Klein tower with spacing $12H$, and no uncertainty bound $p_\mathrm{h} \gtrsim 6H$ on the momentum along the hidden direction. The masses that x_0 produces are of a different kind: a gap above the massless zero modes of the massless bulk fields, set by the warp ($3H$ for scalars, $2H$ for gauge fields; a massless bulk fermion has none), and for dirac16complex a continuum above its bulk mass m_Ψ with eight states bound to the fold at $\omega_\mathrm{b} = \sqrt{m_\Psi^2 - q_\mathrm{b}^2}$ (q_b the inverse width of their profile along the hidden direction). The bound states obey the ordinary uncertainty relation; they are lighter the narrower they are because a narrower state is held by a stronger attraction of the fold, not because confinement costs energy (DERIVED).
2. **Dark matter.** The bound particles χ of dirac16complex cannot decay into the Standard Model alone (whether they are stable against the negative-energy states of their indefinite charge is open), gravitate on the fold like matter of the fold, form a pressureless gas in the dilute limit, and are a thermal relic with the measured density if they annihilate with $\langle \sigma_\mathrm{ann} \mathsf{v} \rangle = 2.2 \times 10^{-26}$ cm^3/s (in the normalisation in which their number density counts all eight states of the bound level; one χ with one antiparticle annihilates four times faster). Direct detection by the LUX-ZEPLIN experiment excludes this through the quark portals, allows it through the lepton portals above about 1.5 TeV, and requires the binding to be weak ($q_\mathrm{b} \lesssim 0.03\,\omega_\mathrm{b}$ at a few TeV); the right-handed neutrinos must not be thermalised. dirac16complex can be the dark matter in this window (DERIVED, ESTIMATE; OPEN: the stability against the negative-energy states).
3. **Dark energy.** No: the gas of bound states has a positive pressure that vanishes in the dilute limit, never a negative one. The dark energy of UM-3 is the vacuum energy of the fold, a free parameter whose pressure is minus its energy density ($w = -1$, w the ratio of pressure to energy density); it fits the cosmic microwave background (CMB) but keeps the tension of the cosmological-constant model with the baryon acoustic oscillations (BAO) of DESI DR2, 2.3 standard deviations in DESI's own analysis (DERIVED: $w = -1$; STANDARD: the tension).
4. **What was left out.** UM-3 had to add right-handed quarks and Yukawa couplings, three right-handed neutrinos (accepted by the author), a thin fluid and an Einstein-Hilbert term on the fold, a medium in the bulk that violates the null energy condition and must respond to the fold, the gauge field that the author's ad hoc Higgs vacuum needs (the doublet $(\mathcal{A}, v + h)/\sqrt{2}$ with the decaying upper component $\mathcal{A} = v_\mathcal{A} \mathrm{e}^{-2a_{40}x_4}$), and it found several open problems (the Krein norm, the extra times, Lorentz violation in the bulk). The last section lists them.

2 The x_0 mass mechanism

2.1 The period of x_0 and the proper length of the hidden direction

$$
\begin{aligned}
g_{MN} &= \mathrm{diag}\Big(\cot^2 z,\ \mathrm{e}^{2a_4}(\sin^2 z)^{1/6},\ \mathrm{e}^{2a_4}(\sin^2 z)^{1/6},\ \mathrm{e}^{2a_4}(\sin^2 z)^{1/6},\ -1,\\
&\qquad -\mathrm{e}^{-2a_4}b_4^2(\sin^2 z)^{1/6},\ -\mathrm{e}^{-2a_4}b_4^2(\sin^2 z)^{1/6},\ -\mathrm{e}^{-2a_4}b_4^2(\sin^2 z)^{1/6}\Big), \qquad z = 6Hx_0,\\
\sigma &= \mathrm{sgn}\Big(\frac{\pi}{2} - z\Big)\frac{\ln\sin z}{6H}, \qquad \mathrm{d}\sigma = |\cot z|\,\mathrm{d}x_0,\\
g_{MN}\,\mathrm{d}X^M\mathrm{d}X^N &= \mathrm{d}\sigma^2 + \mathrm{e}^{-2H|\sigma|}\Big[\mathrm{e}^{2a_4}\big(\mathrm{d}x_1^2 + \mathrm{d}x_2^2 + \mathrm{d}x_3^2\big) - \mathrm{e}^{-2a_4}b_4^2\big(\mathrm{d}x_5^2 + \mathrm{d}x_6^2 + \mathrm{d}x_7^2\big)\Big] - \mathrm{d}x_4^2
\end{aligned}
\tag{FE-27}
$$

Symbols in (FE-27).

- g_{MN}: The metric tensor of the eight-dimensional spacetime (signature $+,+,+,+,-,-,-,-$ in the order $x_0,\ldots,x_7$): $g_{MN}\,\mathrm{d}X^M\mathrm{d}X^N$ is the squared interval, g^{MN} its inverse, $|g|$ the absolute value of its determinant ($\sqrt{|g|}\,\mathrm{d}^8X$ is the proper volume element). Unit: dimensionless. Status: AUTHOR.
- M: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- N: A coordinate index of eight dimensions (also $N_1, N_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- diag: A diagonal matrix with the listed entries on its diagonal (in the order of the coordinates $x_0, x_1, \ldots, x_7$ for a tensor of eight dimensions, $x_1, \ldots, x_7$ for a tensor of the fold, $x_1, \ldots, x_4$ for a four-dimensional tensor). Status: STANDARD.
- z: The angle variable of the hidden direction, $z = 6Hx_0$; one period of x_0 is $0 < z < \pi$, the fold is $z = \pi/2$. Unit: dimensionless. Status: AUTHOR.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- a_4: The author's metric function of the time x_4 that scales ordinary 3-space: on the fold e^{a_4} is the scale factor of cosmology and a_4' the Hubble rate. Unit: dimensionless. Status: AUTHOR.
- b_4: The author's metric function of the time x_4 in the extra-time components, $b_4 > 0$, never identically 1 (the author's order of 2026-10-24); the scale factor of the extra times is $\mathrm{e}^{-a_4}b_4$ times the warp. Unit: dimensionless. Status: AUTHOR.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings and the cosmological constant $\Lambda = -10H^2$; in UM-3 a free parameter. Unit: GeV (1/length). Status: AUTHOR.
- x_0: The hidden space-like coordinate (listed first); the metric depends on it through $z = 6Hx_0$, periodically with the period $\pi/(6H)$. Unit: length. Status: AUTHOR.
- σ: The proper coordinate of the hidden direction: $\mathrm{d}\sigma = |\cot z|\,\mathrm{d}x_0$, $\sigma = \mathrm{sgn}(\pi/2 - z)\ln(\sin z)/(6H)$ on one period $0 < z < \pi$; $\sigma = 0$ is the fold $z = \pi/2$, $\sigma \to \mp\infty$ the tips $z \to 0, \pi$; one period of x_0 is the whole line $-\infty < \sigma < \infty$. Unit: length. Status: DERIVED.
- sgn: The sign function: $\mathrm{sgn}(y) = +1$ for $y > 0$, -1 for $y < 0$. Status: STANDARD.
- π: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, d^nX the volume element of n variables. Status: STANDARD.
- $X = \{x_0, \ldots, x_7\}$: The author's coordinates of the eight-dimensional spacetime, $X = \{x_0, x_1, \ldots, x_7\}$ with the hidden direction first (Dirac_Einstein-Lovelock_4+4_Nash.nb, section DEFCHART); X^M is the coordinate with the index M and $\mathrm{d}^8X = \mathrm{d}x_0\,\mathrm{d}x_1\cdots\mathrm{d}x_7$ the coordinate volume element. Unit: length (each coordinate). Status: AUTHOR.
- x_1: The first coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.
- x_2: The second coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.
- x_3: The third coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.
- x_4: The time ($x_4 \geq 0$ in the ad hoc vacuum); a prime on a function of x_4 alone is $\mathrm{d}/\mathrm{d}x_4$. Unit: length (= time, $c = 1$). Status: AUTHOR.
- x_5: The first of the three extra time-like coordinates. Unit: length. Status: AUTHOR.
- x_6: The second of the three extra time-like coordinates. Unit: length. Status: AUTHOR.
- x_7: The third of the three extra time-like coordinates. Unit: length. Status: AUTHOR.

The author's metric and its proper chart. The metric depends on x_0 only through $z = 6Hx_0$, so its coefficients repeat with the coordinate period $\pi/(6H)$: one period is $0 < z < \pi$, with the fold at $z = \pi/2$ and the tips at $z = 0$ and $z = \pi$, where $g_{x_0x_0} = \cot^2 z$ diverges and the warp $(\sin^2 z)^{1/6}$ vanishes. Status: AUTHOR (the metric), DERIVED (the chart).

$$\int_0^{\pi/(6H)} \sqrt{\cot^2(6Hx_0)}\,\mathrm{d}x_0 = \frac{1}{6H}\int_0^{\pi} |\cot z|\,\mathrm{d}z = \infty, \qquad \sigma(z) = \operatorname{sgn}\Big(\frac{\pi}{2} - z\Big)\frac{\ln\sin z}{6H} \to \mp\infty \quad (z \to 0,\ \pi) \tag{A-1}$$

Symbols in (A-1).

- x_0: The hidden space-like coordinate (listed first); the metric depends on it through $z = 6Hx_0$, periodically with the period $\pi/(6H)$. Unit: length. Status: AUTHOR.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings and the cosmological constant $\Lambda = -10H^2$; in UM-3 a free parameter. Unit: GeV (1/length). Status: AUTHOR.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, d^nX the volume element of n variables. Status: STANDARD.
- z: The angle variable of the hidden direction, $z = 6Hx_0$; one period of x_0 is $0 < z < \pi$, the fold is $z = \pi/2$. Unit: dimensionless. Status: AUTHOR.
- π: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.
- ∞: Infinity: as a limit of an integral or a sum it means that the range has no bound. Status: STANDARD.
- σ: The proper coordinate of the hidden direction: $\mathrm{d}\sigma = |\cot z|\,\mathrm{d}x_0$, $\sigma = \operatorname{sgn}(\pi/2 - z)\ln(\sin z)/(6H)$ on one period $0 < z < \pi$; $\sigma = 0$ is the fold $z = \pi/2$, $\sigma \to \mp\infty$ the tips $z \to 0, \pi$; one period of x_0 is the whole line $-\infty < \sigma < \infty$. Unit: length. Status: DERIVED.
- sgn: The sign function: $\operatorname{sgn}(y) = +1$ for $y > 0$, -1 for $y < 0$. Status: STANDARD.

The proper length of one period of x_0 is infinite: $|\cot z|$ diverges at the tips like $1/z$, so its integral diverges logarithmically, and the proper coordinate σ runs over the whole line. The hidden direction is not a circle of circumference $\pi/(6H)$ but an infinite line, warped by $\mathrm{e}^{-H|\sigma|}$ away from the fold; the tips are infinitely far away (DERIVED, `um3_fold.py`, J1).

$$p_{\mathrm{h}} \gtrsim \frac{\pi}{\pi/(6H)} = 6H, \qquad m_{\mathrm{KK}}^2 = m_\Psi^2 + \big(12H\,\mathsf{k}\big)^2 \quad (\mathsf{k} = 0, \pm1, \pm2, \dots)$$
$$\text{(for a compact direction of length } \pi/(6H)\text{, not UM-3)} \tag{A-2}$$

Symbols in (A-2).

- p_{h}: The momentum along the hidden direction of a particle confined to it, if that direction were compact (A-2, not UM-3). Unit: GeV. Status: STANDARD.
- π: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings and the cosmological constant $\Lambda = -10H^2$; in UM-3 a free parameter. Unit: GeV (1/length). Status: AUTHOR.
- m_{KK}: The masses of the tower of modes of a field on a compact hidden direction (Kaluza-Klein masses; A-2, not UM-3). Unit: GeV. Status: STANDARD.
- m_Ψ: The mass parameter of dirac16complex in the bulk. Unit: GeV. Status: RECORD.
- k: An integer that numbers the modes of a compact direction. Unit: dimensionless. Status: STANDARD.

The mass mechanism one would expect if x_0 were a compact direction of proper length $\pi/(6H)$: a particle confined to it has, by the uncertainty relation, a momentum p_{h} along x_0 of at least about π divided by the length, $6H$, and a field periodic in x_0 has a tower of masses m_{KK} with the spacing $12H$ (the Kaluza-Klein masses, numbered by the integer k). Status: STANDARD (for a compact direction). Neither applies to UM-3, because the proper length is infinite (A-1).

2.2 What the hidden direction produces instead

$$u_{\mathrm{s}}'' + \big(\omega^2 - 9H^2\big)u_{\mathrm{s}} + 6H\,\delta_{\mathrm{D}}(\sigma)\,u_{\mathrm{s}} = 0: \qquad \omega = 0 \ \ \text{(zero mode)}, \qquad \omega \geq 3H \ \ \text{(continuum)} \tag{A-3}$$

Symbols in (A-3).

- u_{s}: The rescaled profile of a massless scalar field of the bulk at zero momentum along ordinary space: the field is $\mathrm{e}^{3H|\sigma|}u_{\mathrm{s}}(\sigma)\,\mathrm{e}^{-\mathrm{i}\omega x_4}$. Unit: as the field. Status: DERIVED.
- $'$: The derivative of a function of the time x_4 alone with respect to x_4 ($a_4' = \mathrm{d}a_4/\mathrm{d}x_4$; two primes the second derivative); on a function of σ alone (a profile) the derivative with respect to σ. Status: STANDARD.
- ω: A frequency: the energy of a mode $\propto \mathrm{e}^{-\mathrm{i}\omega x_4}$ (ω_o of the orbital o). Unit: GeV. Status: STANDARD.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings and the cosmological constant $\Lambda = -10H^2$; in UM-3 a free parameter. Unit: GeV (1/length). Status: AUTHOR.
- δ_{D}: The Dirac delta function: $\int \mathrm{d}\sigma\,\delta_{\mathrm{D}}(\sigma)F(\sigma) = F(0)$. Unit: 1/length (for $\delta_{\mathrm{D}}(\sigma)$). Status: STANDARD.
- σ: The proper coordinate of the hidden direction: $\mathrm{d}\sigma = |\cot z|\,\mathrm{d}x_0$, $\sigma = \mathrm{sgn}(\pi/2 - z)\ln(\sin z)/(6H)$ on one period $0 < z < \pi$; $\sigma = 0$ is the fold $z = \pi/2$, $\sigma \to \mp\infty$ the tips $z \to 0, \pi$; one period of x_0 is the whole line $-\infty < \sigma < \infty$. Unit: length. Status: DERIVED.

A massless field of the bulk (here a scalar at zero momentum along ordinary space, its profile written as $\mathrm{e}^{3H|\sigma|}u_{\mathrm{s}}(\sigma)$): the warp turns its mode equation into a Schrodinger problem with a constant potential $9H^2$ and an attractive delta function at the fold, which binds exactly one state, the massless zero mode; above it the continuum starts at the gap $3H$. A gauge field of the bulk has the same structure with the potential $4H^2$ and the gap $2H$ (its profile written as $\mathrm{e}^{2H|\sigma|}u$), and a massless fermion of the bulk has no gap: once $\mathrm{e}^{3H|\sigma|}$ is taken out, its equation at zero momentum is that of flat space. These gaps are the masses that the geometry of x_0 gives the excitations of bulk fields: they are set by H, as the author's mechanism suggests, but they come from the warp, not from a period, and there is no tower (the spectrum above the gap is continuous). Status: DERIVED (`um3_bulk.py`, B1, for the scalar; the same transformation for the gauge field and the fermion).

$$\chi' = \mathbb{K}\,\chi, \qquad \mathbb{K} = \gamma^{(0)}\big(m_\Psi + \mathrm{i}\omega\gamma^{(4)}\big), \qquad \mathbb{K}^2 = \big(m_\Psi^2 - \omega^2\big)\mathbf{1}, \qquad \chi \propto \mathrm{e}^{\pm q\sigma}, \qquad q = \sqrt{m_\Psi^2 - \omega^2} \tag{FE-37}$$

Symbols in (FE-37).

- $\chi(\sigma)$: The profile of a mode of dirac16complex along the hidden direction, $\Psi = \mathrm{e}^{3H|\sigma|}\chi(\sigma)\mathrm{e}^{\mathrm{i}(\vec{k}\cdot\vec{x} - \omega x_4)}$ (a 16-component spinor function of σ); as a four-dimensional field, χ denotes the particle of a bound state. Unit: GeV^2 (with the V_* normalisation). Status: DERIVED.
- $'$: The derivative of a function of the time x_4 alone with respect to x_4 ($a_4' = \mathrm{d}a_4/\mathrm{d}x_4$; two primes the second derivative); on a function of σ alone (a profile) the derivative with respect to σ. Status: STANDARD.
- $\mathbb{K}$: The transfer matrix of the profile equation at zero momentum, $\chi' = \mathbb{K}\chi$. Unit: GeV. Status: DERIVED.
- γ^M, $\gamma^{(a)}$: The Dirac matrices of dirac16complex along the coordinates, $\gamma^M = \mathbf{e}_a{}^M T^a$, $\gamma_M = g_{MN}\gamma^N$; $\gamma^{(a)} = T^a$ along the frame direction (a) ($\gamma^{(0)}$ along σ, $\gamma^{(4)}$ along the time). Unit: dimensionless. Status: AUTHOR.
- m_Ψ: The mass parameter of dirac16complex in the bulk. Unit: GeV. Status: RECORD.
- i: The imaginary unit, $\mathrm{i}^2 = -1$. Unit: dimensionless. Status: STANDARD.
- ω: A frequency: the energy of a mode $\propto \mathrm{e}^{-\mathrm{i}\omega x_4}$ (ω_o of the orbital o). Unit: GeV. Status: STANDARD.
- $\mathbf{1}$: The identity matrix of the size fixed by the context (16×16 on dirac16complex, 4×4 on a four-dimensional spinor, 3×3 in generation or flavour space). Unit: dimensionless. Status: STANDARD.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- q: The decay rate of a profile along σ at frequency ω, $q = \sqrt{m_\Psi^2 - \omega^2}$ (imaginary in the continuum). Unit: GeV. Status: DERIVED.
- σ: The proper coordinate of the hidden direction: $\mathrm{d}\sigma = |\cot z|\,\mathrm{d}x_0$, $\sigma = \mathrm{sgn}(\pi/2 - z)\ln(\sin z)/(6H)$ on one period $0 < z < \pi$; $\sigma = 0$ is the fold $z = \pi/2$, $\sigma \to \mp\infty$ the tips $z \to 0, \pi$; one period of x_0 is the whole line $-\infty < \sigma < \infty$. Unit: length. Status: DERIVED.

$$q_{\mathrm{b}} = -m_\Psi\left(\mu_{\mathrm{L}} + \mu_{\mathrm{R}}\right)\frac{\tanh\varsigma}{2\varsigma}, \qquad \varsigma = \sqrt{\mu_{\mathrm{L}}\mu_{\mathrm{R}}}, \qquad \omega_{\mathrm{b}} = \sqrt{m_\Psi^2 - q_{\mathrm{b}}^2}, \qquad |\chi_{\mathrm{b}}(\sigma)| \propto \mathrm{e}^{-q_{\mathrm{b}}|\sigma|} \tag{FE-40}$$

Symbols in (FE-40).

- q_{b}: The inverse width of a state of dirac16complex bound to the fold: its profile falls as $\mathrm{e}^{-q_{\mathrm{b}}|\sigma|}$. Unit: GeV. Status: DERIVED.
- m_Ψ: The mass parameter of dirac16complex in the bulk. Unit: GeV. Status: RECORD.
- μ_X, μ_{L}, μ_{R}: The dimensionless strengths of the fold for the left-handed and right-handed halves of dirac16complex, set by the mean fields of the Standard Model at the fold: the jump of the profile across the

fold is the exponential of $\gamma^{(0)}(\mu_{\rm L}P_{\rm L}+\mu_{\rm R}P_{\rm R})$; negative values attract. Unit: dimensionless. Status: DERIVED.
- ς: The auxiliary variable of the bound-state condition, $\varsigma=\sqrt{\mu_{\rm L}\mu_{\rm R}}$ ($\varsigma_-=\sqrt{-\mu_{\rm L}\mu_{\rm R}}$ when $\mu_{\rm L}\mu_{\rm R}<0$). Unit: dimensionless. Status: DERIVED.
- $\omega_{\rm b}$: The energy of a state of dirac16complex bound to the fold at zero momentum: the four-dimensional mass of the particle χ. Unit: GeV. Status: DERIVED.
- $\chi_{\rm b}$: The profile of a state of dirac16complex bound to the fold. Unit: GeV^2. Status: DERIVED.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- σ: The proper coordinate of the hidden direction: $\mathrm{d}\sigma=|\cot z|\,\mathrm{d}x_0$, $\sigma=\mathrm{sgn}(\pi/2-z)\ln(\sin z)/(6H)$ on one period $0<z<\pi$; $\sigma=0$ is the fold $z=\pi/2$, $\sigma\to\mp\infty$ the tips $z\to0,\pi$; one period of x_0 is the whole line $-\infty<\sigma<\infty$. Unit: length. Status: DERIVED.

dirac16complex on the static block (the late-time state of the author's metric in which a_4 and b_4 are constant in the bulk, so that the bulk does not change with time): the continuum of its bulk modes starts at its bulk mass m_Ψ (FE-37), and the fold, if it is attractive, binds eight states below it (FE-40), with the rest energy $\omega_{\rm b}$ seen on the fold and the profile $\mathrm{e}^{-q_{\rm b}|\sigma|}$. Status: DERIVED (`um3_spectra.py`, S1 to S4, with the author's exact 16×16 matrices).

$$\ell_{\rm b}=\sqrt{\langle\sigma^2\rangle}=\frac{1}{\sqrt{2}\,q_{\rm b}},\qquad m_\Psi-\omega_{\rm b}=m_\Psi-\sqrt{m_\Psi^2-q_{\rm b}^2}\approx\frac{q_{\rm b}^2}{2m_\Psi}=\frac{1}{4m_\Psi\,\ell_{\rm b}^2}\tag{A-4}$$

Symbols in (A-4).

- $\ell_{\rm b}$: The width of a bound state of dirac16complex along the hidden direction: the root mean square of σ in its density $\mathrm{e}^{-2q_{\rm b}|\sigma|}$. Unit: length. Value: $1/(\sqrt{2}\,q_{\rm b})$. Status: DERIVED.
- $\langle\cdots\rangle$: The mean (expectation) value of an operator in the state of the medium (the vacuum, a plasma, the Kohn-Sham state). Status: STANDARD.
- σ: The proper coordinate of the hidden direction: $\mathrm{d}\sigma=|\cot z|\,\mathrm{d}x_0$, $\sigma=\mathrm{sgn}(\pi/2-z)\ln(\sin z)/(6H)$ on one period $0<z<\pi$; $\sigma=0$ is the fold $z=\pi/2$, $\sigma\to\mp\infty$ the tips $z\to0,\pi$; one period of x_0 is the whole line $-\infty<\sigma<\infty$. Unit: length. Status: DERIVED.
- $q_{\rm b}$: The inverse width of a state of dirac16complex bound to the fold: its profile falls as $\mathrm{e}^{-q_{\rm b}|\sigma|}$. Unit: GeV. Status: DERIVED.
- m_Ψ: The mass parameter of dirac16complex in the bulk. Unit: GeV. Status: RECORD.
- $\omega_{\rm b}$: The energy of a state of dirac16complex bound to the fold at zero momentum: the four-dimensional mass of the particle χ. Unit: GeV. Status: DERIVED.

The width and the mass of the bound states: their width $\ell_{\rm b}$ along the hidden direction (the root mean square of σ in the density $\mathrm{e}^{-2q_{\rm b}|\sigma|}$) and their mass deficit below the continuum (for weak binding). The states obey the ordinary uncertainty relation: the mean square of the momentum along σ in the profile $\mathrm{e}^{-q_{\rm b}|\sigma|}$ is $q_{\rm b}^2$, so that $\langle p_\sigma^2\rangle\langle\sigma^2\rangle=1/2$. The narrower the state, the larger the deficit, because a narrower state is held by a stronger attraction of the fold: the energy falls not because the particle is confined (in the compact case confinement raises it) but because the fold binds it more strongly; outside the fold the profile is an evanescent wave, $p_\sigma^2=-q_{\rm b}^2$ in $\omega^2=m_\Psi^2+p_\sigma^2$. The x_0 mass of dirac16complex is therefore a binding mass: $\omega_{\rm b}=\sqrt{m_\Psi^2-q_{\rm b}^2}$, set by the bulk mass and by the strengths of the fold terms (the mean fields of the Standard Model at the fold), not by the period of x_0. Status: DERIVED.

2.3 The answer

The period $\pi/(6H)$ of the coordinate x_0 does not produce a mass by the uncertainty relation, because one period is an infinite proper distance. The hidden direction produces masses in two other ways: the warp gives the massless bulk scalars a gap $3H$ and the bulk gauge fields a gap $2H$ above their zero modes (a mass scale set by H, without a tower; a massless bulk fermion has no gap), and the fold binds dirac16complex in eight states of mass $\omega_{\rm b}=\sqrt{m_\Psi^2-q_{\rm b}^2}$, which are lighter the narrower they are, because a narrower state is bound more strongly. Status: DERIVED.

3 dirac16complex as dark matter

3.1 Stability

$$\mathcal{L}_{\mathrm{P}} = -\,c_{\Phi}\,\big(\Phi^{\dagger}\Phi\big)\,\bar{\Psi}P_{\mathrm{L}}\Psi - \sum_{X}\Big[\frac{1}{4}\big(c_X^{G}\,G^{A}_{\mu\nu}G^{A\mu\nu} + c_X^{W}\,W^{I}_{\mu\nu}W^{I\mu\nu} + c_X^{B}\,B_{\mu\nu}B^{\mu\nu}\big)\,\bar{\Psi}P_X\Psi + \frac{d_X}{2}\,B_{\mu\nu}\,\mathcal{D}_X^{\mu\nu} + \sum_{f} c_X^{f}\,J_f^{\mu}\,j_{X\mu}\Big], \tag{IN-16}$$

$$\mathcal{D}_X^{\mu\nu} = \mathrm{i}\,\bar{\Psi}\,\Sigma^{\mu\nu}P_X\Psi, \qquad j_X^{\mu} = \mathrm{i}\big(\bar{\Psi}P_X D^{\mu}\Psi - (D^{\mu}\bar{\Psi})P_X\Psi\big), \qquad J_f^{\mu} = \bar{\psi}_f\,\hat{\gamma}^{\mu}\psi_f$$

Symbols in (IN-16).

- $\mathcal{L}_{\mathrm{P}}$: The Lagrangian of the portals of dirac16complex at the fold. Unit: GeV^4. Status: INVENTED.
- c_{Φ}: The Higgs-portal coupling of the left-handed half of dirac16complex (the right-handed half has none). Unit: GeV^{-5}. Status: INVENTED.
- Φ: The Higgs doublet (two complex components, hypercharge 1/2), a field of the fold. Unit: GeV. Status: AUTHOR.
- †: Hermitian conjugation: the complex conjugate of the transpose of a matrix or spinor (for field operators the adjoint operator). Status: STANDARD.
- Ψ, $\bar{\Psi}$: dirac16complex: the author's 16-component complex spinor field of eight dimensions; $\bar{\Psi} = \Psi^{\dagger}\sigma_{16}$ is its Dirac adjoint; in a term of the fold, Ψ is its value at $\sigma = 0$. Unit: $\mathrm{GeV}^{7/2}$. Status: AUTHOR.
- P_{L}: The author's left-handed chiral projector, $P_{\mathrm{L}} = (\mathbf{1} - \Gamma_8)/2$. Unit: dimensionless. Status: AUTHOR.
- X: The label of a half of dirac16complex: $X \in \{\mathrm{L}, \mathrm{R}\}$ (left-handed or right-handed half); a sum over X runs over both. Status: STANDARD.
- c_X^{G}: The gluon-kinetic portal coupling of the half X of dirac16complex. Unit: GeV^{-7}. Status: INVENTED.
- G_{μ}^{A}, $G_{\mu\nu}^{A}$: The gluon field of $SU(3)_C$ on the fold ($A = 1, \ldots, 8$) and its field strength. Unit: GeV (field), GeV^2 (field strength). Status: AUTHOR.
- A: A colour (adjoint SU(3)) index running over $1, \ldots, 8$; a repeated index is summed. Status: STANDARD.
- μ: A four-dimensional coordinate index running over the observable coordinates x_1, x_2, x_3, x_4 of the fold (signature $+,+,+,-$); a repeated index is summed. Status: STANDARD.
- ν: A four-dimensional coordinate index (as μ). Status: STANDARD.
- c_X^{W}: The W-kinetic portal coupling of the half X. Unit: GeV^{-7}. Status: INVENTED.
- W_{μ}^{I}, $W_{\mu\nu}^{I}$: The gauge field of $SU(2)_L$ on the fold, $W_{\mu}^{1}, W_{\mu}^{2}, W_{\mu}^{3}$, and its field strength. Unit: GeV (field), GeV^2 (field strength). Status: AUTHOR.
- I: A weak-isospin (adjoint SU(2)) index running over $1, 2, 3$; a repeated index is summed. Status: STANDARD.
- c_X^{B}: The B-kinetic portal coupling of the half X. Unit: GeV^{-7}. Status: INVENTED.
- B_{μ}, $B_{\mu\nu}$: The hypercharge gauge field of $U(1)_Y$ on the fold and its field strength. Unit: GeV (field), GeV^2 (field strength). Status: AUTHOR.
- P_X: The chiral projector of the half X: $P_X = P_{\mathrm{L}}$ for $X = \mathrm{L}$, P_{R} for $X = \mathrm{R}$. Unit: dimensionless. Status: AUTHOR.
- d_X: The dipole-portal coupling of the half X of dirac16complex with the hypercharge field strength. Unit: GeV^{-5}. Status: INVENTED.
- $\mathcal{D}_X^{\mu\nu}$: The dipole (antisymmetric tensor) density of the half X of dirac16complex at the fold, $\mathcal{D}_X^{\mu\nu} = \mathrm{i}\bar{\Psi}\Sigma^{\mu\nu}P_X\Psi$ (real, DERIVED). Unit: GeV^7. Status: DERIVED.
- f: The label of a kind of Standard-Model multiplet: $f \in \{Q, u, d, \ell, e, \nu\}$ (quark doublets, up-quark singlets, down-quark singlets, lepton doublets, charged-lepton singlets, right-handed neutrinos); a sum over f runs over the six. Status: STANDARD.
- c_X^{f}: The current-portal coupling of the half X of dirac16complex with the fermion-number current of the multiplets of kind f (flavour blind); c_X^{ℓ} and c_X^{ν} are those of the lepton doublets and of the right-handed neutrinos. Unit: GeV^{-7}. Status: INVENTED.
- J_f^{μ}: The fermion-number current of the Standard-Model multiplets of kind f, $J_f^{\mu} = \bar{\psi}_f\hat{\gamma}^{\mu}\psi_f$ summed over the three generations and the gauge components; with other labels the currents that are the sources of the gauge fields. Unit: GeV^3. Status: STANDARD.
- j_X^{μ}: The convective current of the half X of dirac16complex, $j_X^{\mu} = \mathrm{i}(\bar{\Psi}P_X D^{\mu}\Psi - (D^{\mu}\bar{\Psi})P_X\Psi)$ (real and joining the half with itself, DERIVED). Unit: GeV^8. Status: DERIVED.
- i: The imaginary unit, $\mathrm{i}^2 = -1$. Unit: dimensionless. Status: STANDARD.

- Σ^{ab}, $\Sigma^{\mu\nu}$: The generators of Spin(4,4) on dirac16complex, $\Sigma^{ab} = \frac{1}{4}[T^a, T^b]$ (the scaled commutators of the author's matrices); $\Sigma^{\mu\nu} = \mathbf{e}_a{}^\mu \mathbf{e}_b{}^\nu \Sigma^{ab}$ along the fold. Unit: dimensionless. Status: AUTHOR.
- D_M, D_μ: The covariant derivative: on dirac16complex $D_M = \partial_M + \Omega_M$ (with $D_M \bar\Psi = \partial_M \bar\Psi - \bar\Psi \Omega_M$); on the fields of the fold the gauge and spin covariant derivative D_μ (the partial derivative plus the spin connection of the fold and the gauge fields times their couplings and generators); $D^\mu = \hat h^{\mu\nu} D_\nu$; on an adjoint field $(D_\mu X)^A = \partial_\mu X^A + g_3 f^{ABC} G^B_\mu X^C$ (likewise for $SU(2)$ with $g_2 \epsilon^{IJK}$). Unit: 1/length. Status: STANDARD.
- ψ, $\bar\psi$: A Standard-Model fermion field (a four-dimensional Weyl field of the fold with three generations, written as a four-component spinor of fixed chirality); $\bar\psi = \psi^\dagger \beta$; ψ_f the fields of the kind f. Unit: $\mathrm{GeV}^{3/2}$. Status: STANDARD.
- $\hat\gamma^\mu$: The four-dimensional complex 4×4 Dirac matrices of the fold, $\hat\gamma^\mu = \hat{\mathbf{e}}_a{}^\mu \hat\gamma^{(a)}$, $\{\hat\gamma^\mu, \hat\gamma^\nu\} = -2\hat h^{\mu\nu}\mathbf{1}$; $\hat\gamma_\mu = \hat h_{\mu\nu}\hat\gamma^\nu$. Unit: dimensionless. Status: STANDARD.

$$Q_\Psi = \int \mathrm{d}\sigma\, \mathrm{d}^3x\, \sqrt{|g|}\, \Psi^\dagger \mathfrak{B} \Psi, \qquad \mathfrak{B} = \mathrm{i}\,\sigma_{16}\,\gamma^{(4)}, \qquad \mathfrak{B}^\dagger = \mathfrak{B}, \qquad \mathfrak{B}^2 = \mathbf{1}, \qquad E_o = -\omega_o\, \mathcal{Q}_o \tag{FE-43}$$

Symbols in (FE-43).

- Q_Ψ: The conserved charge of dirac16complex (the Noether charge of a change of the phase of Ψ by a constant), a Krein form (indefinite). Unit: dimensionless. Status: DERIVED.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, $\mathrm{d}^n X$ the volume element of n variables. Status: STANDARD.
- σ: The proper coordinate of the hidden direction: $\mathrm{d}\sigma = |\cot z|\,\mathrm{d}x_0$, $\sigma = \mathrm{sgn}(\pi/2 - z)\ln(\sin z)/(6H)$ on one period $0 < z < \pi$; $\sigma = 0$ is the fold $z = \pi/2$, $\sigma \to \mp\infty$ the tips $z \to 0, \pi$; one period of x_0 is the whole line $-\infty < \sigma < \infty$. Unit: length. Status: DERIVED.
- x: The coordinates of the fold: x_i $(i = 1, 2, 3)$ of ordinary space, x_4 the time, x_r $(r = 5, 6, 7)$ the extra times; a field of the fold is a function of the four observable coordinates $x = (x_1, x_2, x_3, x_4)$, and $\mathrm{d}^4x = \mathrm{d}x_1\,\mathrm{d}x_2\,\mathrm{d}x_3\,\mathrm{d}x_4$. Unit: length (each coordinate). Status: AUTHOR.
- g_{MN}: The metric tensor of the eight-dimensional spacetime (signature $+,+,+,+,-,-,-,-$ in the order $x_0, \ldots, x_7$): $g_{MN}\,\mathrm{d}X^M \mathrm{d}X^N$ is the squared interval, g^{MN} its inverse, $|g|$ the absolute value of its determinant ($\sqrt{|g|}\,\mathrm{d}^8X$ is the proper volume element). Unit: dimensionless. Status: AUTHOR.
- Ψ, $\bar\Psi$: dirac16complex: the author's 16-component complex spinor field of eight dimensions; $\bar\Psi = \Psi^\dagger \sigma_{16}$ is its Dirac adjoint; in a term of the fold, Ψ is its value at $\sigma = 0$. Unit: $\mathrm{GeV}^{7/2}$. Status: AUTHOR.
- †: Hermitian conjugation: the complex conjugate of the transpose of a matrix or spinor (for field operators the adjoint operator). Status: STANDARD.
- $\mathfrak{B}$: The Krein operator, $\mathfrak{B} = \mathrm{i}\sigma_{16}\gamma^{(4)}$: a Hermitian involution with the signature (8, 8); $\Psi^\dagger \mathfrak{B} \Psi$ is the charge density. Unit: dimensionless. Status: DERIVED.
- i: The imaginary unit, $\mathrm{i}^2 = -1$. Unit: dimensionless. Status: STANDARD.
- σ_{16}: The matrix of the Dirac adjoint, $\sigma_{16} = T_0 T_1 T_2 T_3$, real and symmetric; $\sigma_{16} T_a$ is real and antisymmetric. Unit: dimensionless. Status: AUTHOR.
- γ^M, $\gamma^{(a)}$: The Dirac matrices of dirac16complex along the coordinates, $\gamma^M = \mathbf{e}_a{}^M T^a$, $\gamma_M = g_{MN}\gamma^N$; $\gamma^{(a)} = T^a$ along the frame direction (a) ($\gamma^{(0)}$ along σ, $\gamma^{(4)}$ along the time). Unit: dimensionless. Status: AUTHOR.
- $\mathbf{1}$: The identity matrix of the size fixed by the context (16×16 on dirac16complex, 4×4 on a four-dimensional spinor, 3×3 in generation or flavour space). Unit: dimensionless. Status: STANDARD.
- E: An energy: E_o the energy of the Kohn-Sham orbital o; in the neutrino sector the energy of a neutrino. Unit: GeV. Status: STANDARD.
- o: The label of a Kohn-Sham orbital (a single-particle state of dirac16complex); a sum over o runs over the orbitals. Status: STANDARD.
- ω: A frequency: the energy of a mode $\propto \mathrm{e}^{-\mathrm{i}\omega x_4}$ (ω_o of the orbital o). Unit: GeV. Status: STANDARD.
- $\mathcal{Q}_o$: The charge Q_Ψ of the Kohn-Sham orbital o (positive or negative). Unit: dimensionless. Status: DERIVED.

The portals of dirac16complex to the Standard Model (IN-16) and its conserved charge (FE-43). Every portal term, like the bulk Lagrangian and the self-interaction, contains dirac16complex only in the bilinears $\bar\Psi \cdots \Psi$: the action is invariant under a constant change of the phase of Ψ, and the charge Q_Ψ is conserved. A particle χ (a bound state of dirac16complex) can therefore annihilate only with an antiparticle, and the lightest states of dirac16complex, the bound states of the fold, cannot decay into the Standard Model alone. Conservation of Q_Ψ does not by itself make them stable, because the charge is a Krein form (indefinite): with $E_o = -\omega_o \mathcal{Q}_o$, a particle χ of positive energy (charge -1 at $+\omega_\mathrm{b}$) could turn into a state of negative energy with the same charge (at $-\omega_\mathrm{b}$) and Standard-Model

quanta that carry away $2\omega_\mathrm{b}$. Whether this happens, and how fast, is not settled. Status: DERIVED (no decay into the Standard Model alone, from the form of IN-16; the Krein signature of the charge, `um3_fields.py`, F3, and `um3_spectra.py`, S4), OPEN (the stability against the negative-energy states).

3.2 Gravitation on the fold

$$\kappa \int_{\text{layer}} \mathcal{T}_\Psi{}^a{}_b \, \mathrm{d}\sigma = \frac{\kappa}{\mathcal{Z}\, V_*} \, \mathcal{T}_\chi{}^a{}_b \qquad \left(H \ll q_\mathrm{b}\right) \tag{A-5}$$

Symbols in (A-5).

- κ: The eight-dimensional gravitational coupling, $\kappa = 8\pi G_8$. Unit: GeV^{-6}. Status: AUTHOR.
- $\mathcal{T}_\Psi$: The energy-momentum tensor of dirac16complex, $\mathcal{T}_{MN} = -(2/\sqrt{|g|})\, \delta S_\Psi / \delta g^{MN}$. Unit: GeV^8. Status: STANDARD.
- a: A frame (orthonormal) index running over $0, 1, \ldots, 7$ (frame direction (0) along σ, (1), (2), (3) along ordinary space, (4) along the time, (5), (6), (7) along the extra times); raised and lowered with η_{ab}; on the fold ($a, b \in \{1, \ldots, 7\}$) the frame of the fold; a repeated index is summed. Status: STANDARD.
- b: A frame index (as a). Status: STANDARD.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, $\mathrm{d}^n X$ the volume element of n variables. Status: STANDARD.
- σ: The proper coordinate of the hidden direction: $\mathrm{d}\sigma = |\cot z| \, \mathrm{d}x_0$, $\sigma = \mathrm{sgn}(\pi/2 - z) \ln(\sin z)/(6H)$ on one period $0 < z < \pi$; $\sigma = 0$ is the fold $z = \pi/2$, $\sigma \to \mp\infty$ the tips $z \to 0, \pi$; one period of x_0 is the whole line $-\infty < \sigma < \infty$. Unit: length. Status: DERIVED.
- $\mathcal{Z}$: The fold factor, $\mathcal{Z}(x_4) = V_\mathrm{t}(\mathrm{e}^{-a_4} b_4)^3 / V_*$: the factor by which the size of the extra times multiplies every term of the fold; $\mathcal{Z} = 1$ when the extra times are frozen at their late value. Unit: dimensionless. Value: 1 in the late universe. Status: DERIVED.
- V_*: The physical volume of the extra times x_5, x_6, x_7 on the fold in the late universe, $V_* = V_\mathrm{t}(\mathrm{e}^{-a_4} b_4)^3$ at a time when $\mathrm{e}^{-a_4} b_4$ is constant; it converts the eight-dimensional normalisation into the four-dimensional one. Unit: length^3 (GeV^{-3}). Status: ASSUMED.
- $\mathcal{T}_\chi{}^a{}_b$: The four-dimensional energy-momentum tensor of the gas of the bound particles χ of dirac16complex, per proper volume of ordinary space at the fold (energy density ρ_χ). Unit: GeV^4. Status: DERIVED.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings and the cosmological constant $\Lambda = -10H^2$; in UM-3 a free parameter. Unit: GeV (1/length). Status: AUTHOR.
- q_b: The inverse width of a state of dirac16complex bound to the fold: its profile falls as $\mathrm{e}^{-q_\mathrm{b}|\sigma|}$. Unit: GeV. Status: DERIVED.

The particles χ live in the bulk, in a layer of width $1/q_\mathrm{b}$ around the fold. The junction conditions of the fold integrate the bulk field equations across the fold: the jump of the junction potential across the fold equals κ times the energy-momentum tensor $\mathcal{S}$ of everything that lives on the fold. If the layer is thin compared with the scale $1/H$ on which the metric varies, integrating across the whole layer gives that jump the additional source A-5, where $\mathcal{T}_\chi$ is the four-dimensional energy-momentum tensor of the gas of χ (per proper volume of ordinary space) and $\mathcal{Z}V_*$ the proper volume of the extra times at the fold ($\mathcal{Z} = 1$ with frozen extra times, that is when the size $\mathrm{e}^{-a_4} b_4$ of the extra times no longer changes). This is exactly the form in which the matter of the fold enters ($\mathcal{S} = \mathcal{T}_\mathrm{SM}/V_* + \ldots$, the energy-momentum tensor of the Standard Model divided by the volume of the extra times): the particles χ gravitate on the fold like matter of the fold, with the same Newton constant, and enter the Friedmann equation of FE-59 as ρ_χ. The corrections are of relative order H/q_b. Status: DERIVED (from the junction conditions), ESTIMATE (the corrections).

3.3 The equation of state of the Kohn-Sham states

$$\mathsf{E}_\mathrm{int} = \frac{G_\mathrm{KS}}{2} \int_{-\infty}^{\infty} \mathrm{d}\sigma \, \mathrm{e}^{6H|\sigma|} \left(\hat{\rho}^2 - \mathrm{tr}\, \hat{\mathcal{N}}^2 \right), \qquad \epsilon_\mathrm{p} = \omega - \mathsf{E}_\mathrm{int}, \qquad P_\mathrm{KS} = n_\chi \, \mathsf{E}_\mathrm{int}, \qquad w_\chi = \frac{P_\mathrm{KS}}{n_\chi \, \epsilon_\mathrm{p}} \tag{A-8}$$

Symbols in (A-8).

- E_int: The interaction energy per particle of the gas of χ: the mean self-interaction energy $\frac{\lambda_\Psi}{2}(n_\mathrm{s}^2 - \mathrm{tr}\, \mathcal{N}^2)$ integrated across the hidden direction, per particle. Unit: GeV. Status: DERIVED.
- G_KS: The dimensionless strength of the self-interaction of the gas of particles χ in the Kohn-Sham states: the self-coupling λ_Ψ times the number density n_χ of the particles on the fold divided by the volume V_* of the extra

times, $G_{\rm KS} = \lambda_\Psi n_\chi / V_*$ (in units of m_Ψ when numbers are given). Unit: dimensionless. Value: 10^{-6} to 1 in the solutions. Status: DERIVED.

- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, $\mathrm{d}^n X$ the volume element of n variables. Status: STANDARD.
- σ: The proper coordinate of the hidden direction: $\mathrm{d}\sigma = |\cot z|\,\mathrm{d}x_0$, $\sigma = \mathrm{sgn}(\pi/2 - z)\ln(\sin z)/(6H)$ on one period $0 < z < \pi$; $\sigma = 0$ is the fold $z = \pi/2$, $\sigma \to \mp\infty$ the tips $z \to 0, \pi$; one period of x_0 is the whole line $-\infty < \sigma < \infty$. Unit: length. Status: DERIVED.
- ∞: Infinity: as a limit of an integral or a sum it means that the range has no bound. Status: STANDARD.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings and the cosmological constant $\Lambda = -10H^2$; in UM-3 a free parameter. Unit: GeV (1/length). Status: AUTHOR.
- $\hat\rho(\sigma)$: The scalar density of the occupied orbitals, $\hat\rho = \frac{1}{4}\sum_o \bar\chi_o \chi_o$ (per particle); the scalar density of the gas is $n_{\rm s} = (n_\chi/V_*)\,\mathrm{e}^{6H|\sigma|}\hat\rho$. Unit: GeV. Status: DERIVED.
- tr: The trace of a matrix: the sum of its diagonal entries. Status: STANDARD.
- $\hat{\mathcal{N}}(\sigma)$: The one-body density matrix of the occupied orbitals, $\hat{\mathcal{N}} = \frac{1}{4}\sum_o \chi_o \bar\chi_o$ (a 16×16 matrix at each σ, per particle); the density matrix of the gas is $\mathcal{N} = (n_\chi/V_*)\,\mathrm{e}^{6H|\sigma|}\hat{\mathcal{N}}$. Unit: GeV. Status: DERIVED.
- $\epsilon_{\rm p}$: The energy per particle of the gas of χ: the orbital energy minus the interaction energy per particle, which the orbital energy counts twice. Unit: GeV. Status: DERIVED.
- ω: A frequency: the energy of a mode $\propto \mathrm{e}^{-\mathrm{i}\omega x_4}$ (ω_o of the orbital o). Unit: GeV. Status: STANDARD.
- $P_{\rm KS}$: The pressure of the gas of χ along ordinary space in the Kohn-Sham state: the particles are at rest, so it is the mean self-interaction, $P_{\rm KS} = n_\chi \mathsf{E}_{\rm int}$. Unit: GeV^4. Status: DERIVED.
- n_χ: The number density of the dark-matter particles χ (particles plus antiparticles). Unit: GeV^3. Status: DERIVED.
- w_χ: The equation of state of the gas of the particles χ: its pressure divided by its energy density, $w_\chi = P_{\rm KS}/(n_\chi \epsilon_{\rm p})$ (0 for cold dark matter, -1 for a vacuum energy). Unit: dimensionless. Status: DERIVED.

The equation of state of a homogeneous gas of the particles χ at rest on the fold, from its self-consistent Kohn-Sham states. Each particle occupies one of the four positive-energy states of the bound level, with the field $\Psi = \mathrm{e}^{3H|\sigma|}\chi_o(\sigma)\,\mathrm{e}^{-\mathrm{i}\omega x_4}$; the profiles χ_o obey the Dirac equation along the hidden direction with the mean field $G_{\rm KS}\,\mathrm{e}^{6H|\sigma|}(\hat\rho\,\mathbf{1} - \hat{\mathcal{N}})$ of the self-interaction added to the mass, where $\hat\rho$ and $\hat{\mathcal{N}}$ are the scalar density and the density matrix of the profiles, $\mathrm{e}^{6H|\sigma|}$ is the warp of the proper density, and $G_{\rm KS} = \lambda_\Psi n_\chi/V_*$ is the dimensionless strength of the self-interaction (the self-coupling λ_Ψ of dirac16complex times the number density n_χ of the particles on the fold, divided by the volume V_* of the extra times). ω is the orbital energy, $\mathsf{E}_{\rm int}$ the interaction energy per particle, $\epsilon_{\rm p}$ the energy per particle (the orbital energy counts the interaction twice), and $P_{\rm KS}$ the pressure: the particles are at rest and do not depend on the extra times, so only the self-interaction contributes, along ordinary space and along the extra times alike; w_χ is the equation of state, the pressure over the energy density. Status: DERIVED.

The Kohn-Sham equations were solved (in units of m_Ψ, with the Higgs portal of the vacuum, $q_{\rm b} = 0.3$) in three regimes of the bulk scale: $H = 0.05$, where the proper density of the bound state without the self-interaction decays away from the fold; $H = 0.15$, where it grows but the first-order shift of the level is finite; and $H = 0.3$, where both grow. In every regime and at every coupling from $G_{\rm KS} = 10^{-6}$ to 1 the self-interaction is repulsive for $\lambda_\Psi > 0$, the level rises, and the pressure is positive: w_χ takes the values 7.2×10^{-8}, 2.1×10^{-7}, 4.7×10^{-5} at $G_{\rm KS} = 10^{-6}$ and 0.056, 0.137, 0.224 at $G_{\rm KS} = 1$ in the three regimes. Where the proper density would grow, the repulsion makes it saturate at a constant value far from the fold, and the level can rise above the threshold m_Ψ of the free continuum without unbinding (at $G_{\rm KS} = 0.371$ and 0.142 in the first two regimes). At a weakly bound point of the window below ($q_{\rm b} = 0.03$, $H = 0.002$) the self-consistent shift approaches first-order perturbation theory as $G_{\rm KS} \to 0$ and w_χ takes the values 6.2×10^{-7}, 6.2×10^{-6}, 5.8×10^{-5} at $G_{\rm KS} = 10^{-4}$, 10^{-3} and 10^{-2}. Status: DERIVED (`um3_kohnsham.py`, K3, K7, K8, K9; 9 checks).

How dilute is the gas of the universe? With the self-coupling of the natural size $\lambda_\Psi = m_\Psi^{-6}$ and extra times of the size $1/m_\Psi$ ($V_* = m_\Psi^{-3}$), $G_{\rm KS} = n_\chi/m_\Psi^3$; for $m_\Psi = 1.5$ TeV and the density of dark matter near the Sun (0.4 $\mathrm{GeV/cm}^3$, $n_\chi = 2.7\times10^{-4}\ \mathrm{cm}^{-3}$) this is $G_{\rm KS} \approx 6\times10^{-55}$, and the mean density of the universe is smaller still. $G_{\rm KS}$ reaches 10^{-6} only if λ_Ψ/V_* is larger by a factor of about 10^{48}; extra times of the Planck length ($V_* = M_{\rm Pl}^{-3}$) give 5×10^{47}, so $G_{\rm KS} \approx 3\times10^{-7}$. The cosmic gas of χ is therefore deep in the dilute limit, where w_χ is far below the values above: it is cold dark matter. Status: ESTIMATE.

3.4 The relic density, direct detection and the window

$$\mathfrak{g}_\Phi = \frac{\partial\omega_\mathrm{b}}{\partial\mu_\mathrm{L}}\,\frac{c_\Phi}{V_*}, \qquad \mathfrak{g}_G = \sum_X \frac{\partial\omega_\mathrm{b}}{\partial\mu_X}\,\frac{c_X^G}{4V_*}, \qquad \mathfrak{g}_f = \sum_X \frac{\partial\omega_\mathrm{b}}{\partial\mu_X}\,\frac{2\omega_\mathrm{b}\,c_X^f}{V_*}, \qquad \frac{\partial\omega_\mathrm{b}}{\partial\mu_X} = -\frac{q_\mathrm{b}}{\omega_\mathrm{b}}\,\frac{\partial q_\mathrm{b}}{\partial\mu_X},$$
$$\Delta\omega_\mathrm{b} = \mathfrak{g}_\Phi\,\Delta\langle\Phi^\dagger\Phi\rangle + \mathfrak{g}_G\,\Delta\langle G^2\rangle + \sum_f \mathfrak{g}_f\,\Delta n_f, \qquad \mathfrak{g}_{h\chi} = \mathfrak{g}_\Phi\,v = -\frac{m_\Psi^2\,\mu_\mathrm{L}^2}{2\,\omega_\mathrm{b}\,v} \quad (\mu_\mathrm{R}=0) \tag{FE-47}$$

Symbols in (FE-47).

- $\mathfrak{g}_\Phi$, $\mathfrak{g}_G$, $\mathfrak{g}_f$, $\mathfrak{g}_{h\chi}$: The four-dimensional couplings of the bound states χ of dirac16complex to the Standard Model (FE-47): to $\Phi^\dagger\Phi$, to the gluon kinetic density, to the number densities of the fermions of kind f, and the Yukawa coupling of the Higgs boson to χ. Unit: $\mathfrak{g}_\Phi$: GeV^{-1}; $\mathfrak{g}_G$: GeV^{-3}; $\mathfrak{g}_f$: GeV^{-2}; $\mathfrak{g}_{h\chi}$: dimensionless. Status: DERIVED.
- ∂: A partial derivative: ∂_M with respect to the coordinate X^M, ∂_μ with respect to x^μ, ∂_σ with respect to σ, $\partial X/\partial Y$ the derivative of X with respect to Y at fixed other variables. Status: STANDARD.
- ω_b: The energy of a state of dirac16complex bound to the fold at zero momentum: the four-dimensional mass of the particle χ. Unit: GeV. Status: DERIVED.
- μ_X, μ_L, μ_R: The dimensionless strengths of the fold for the left-handed and right-handed halves of dirac16complex, set by the mean fields of the Standard Model at the fold: the jump of the profile across the fold is the exponential of $\gamma^{(0)}(\mu_\mathrm{L}P_\mathrm{L} + \mu_\mathrm{R}P_\mathrm{R})$; negative values attract. Unit: dimensionless. Status: DERIVED.
- c_Φ: The Higgs-portal coupling of the left-handed half of dirac16complex (the right-handed half has none). Unit: GeV^{-5}. Status: INVENTED.
- V_*: The physical volume of the extra times x_5, x_6, x_7 on the fold in the late universe, $V_* = V_\mathrm{t}(\mathrm{e}^{-a_4}b_4)^3$ at a time when $\mathrm{e}^{-a_4}b_4$ is constant; it converts the eight-dimensional normalisation into the four-dimensional one. Unit: length^3 (GeV^{-3}). Status: ASSUMED.
- c_X^G: The gluon-kinetic portal coupling of the half X of dirac16complex. Unit: GeV^{-7}. Status: INVENTED.
- X: The label of a half of dirac16complex: $X \in \{\mathrm{L}, \mathrm{R}\}$ (left-handed or right-handed half); a sum over X runs over both. Status: STANDARD.
- c_X^f: The current-portal coupling of the half X of dirac16complex with the fermion-number current of the multiplets of kind f (flavour blind); c_X^ℓ and c_X^ν are those of the lepton doublets and of the right-handed neutrinos. Unit: GeV^{-7}. Status: INVENTED.
- f: The label of a kind of Standard-Model multiplet: $f \in \{Q, u, d, \ell, e, \nu\}$ (quark doublets, up-quark singlets, down-quark singlets, lepton doublets, charged-lepton singlets, right-handed neutrinos); a sum over f runs over the six. Status: STANDARD.
- q_b: The inverse width of a state of dirac16complex bound to the fold: its profile falls as $\mathrm{e}^{-q_\mathrm{b}|\sigma|}$. Unit: GeV. Status: DERIVED.
- Δ: A change, difference or small variation of the quantity that follows ($\Delta\rho$ a perturbation, Δm^2 a difference of squared masses, ΔN_eff an addition to N_eff). Status: STANDARD.
- Φ: The Higgs doublet (two complex components, hypercharge 1/2), a field of the fold. Unit: GeV. Status: AUTHOR.
- †: Hermitian conjugation: the complex conjugate of the transpose of a matrix or spinor (for field operators the adjoint operator). Status: STANDARD.
- $\langle\cdots\rangle$: The mean (expectation) value of an operator in the state of the medium (the vacuum, a plasma, the Kohn-Sham state). Status: STANDARD.
- $\langle G^2\rangle$: The mean square of the gluon field strength, $\langle G^A_{\mu\nu}G^{A\mu\nu}\rangle$ (the gluon condensate in the QCD vacuum, a thermal value in the plasma). Unit: GeV^4. Status: STANDARD.
- G^A_μ, $G^A_{\mu\nu}$: The gluon field of $SU(3)_C$ on the fold ($A = 1, \ldots, 8$) and its field strength. Unit: GeV (field), GeV^2 (field strength). Status: AUTHOR.
- n_f: The number density of the Standard-Model fermions of kind f (particles minus antiparticles, summed over generations and gauge components), the time component of the mean current $\langle J^\mu_f\rangle$ in the rest frame of the medium. Unit: GeV^3. Status: STANDARD.
- v: The vacuum expectation value of the lower component of the Higgs doublet, $v^2 = -\mu^2/\lambda$. Unit: GeV. Value: 246.22 GeV $= (\sqrt{2}\,G_\mathrm{F})^{-1/2}$ with the Fermi constant $G_\mathrm{F} = 1.1663788 \times 10^{-5}$ GeV^{-2} of the muon lifetime (Particle Data Group). Status: STANDARD.
- m_Ψ: The mass parameter of dirac16complex in the bulk. Unit: GeV. Status: RECORD.

$$\frac{\mathrm{d}n_\chi}{\mathrm{d}x_4} + 3H_4\, n_\chi = -\langle\sigma_{\mathrm{ann}}\mathsf{v}\rangle\big(n_\chi^2 - n_{\chi,\mathrm{eq}}^2\big), \qquad n_{\chi,\mathrm{eq}} = g_\chi\Big(\frac{\omega_\mathrm{b}\mathsf{T}}{2\pi}\Big)^{3/2}\mathrm{e}^{-\omega_\mathrm{b}/\mathsf{T}}, \qquad \Omega_\chi h_{100}^2 \simeq \frac{1.07\times10^9\ \mathrm{GeV}^{-1}\, x_\mathrm{f}}{\sqrt{g_*}\, M_\mathrm{Pl}\,\langle\sigma_{\mathrm{ann}}\mathsf{v}\rangle} \tag{FE-61}$$

Symbols in (FE-61).

- d: The differential: $\mathrm{d}X$ is an infinitesimal change of X, $\mathrm{d}X/\mathrm{d}Y$ the derivative of X with respect to Y, $\mathrm{d}^n X$ the volume element of n variables. Status: STANDARD.
- n_χ: The number density of the dark-matter particles χ (particles plus antiparticles). Unit: GeV^3. Status: DERIVED.
- x_4: The time ($x_4 \geq 0$ in the ad hoc vacuum); a prime on a function of x_4 alone is $\mathrm{d}/\mathrm{d}x_4$. Unit: length (= time, $c = 1$). Status: AUTHOR.
- H_4: The expansion rate (Hubble rate) of the fold, $H_4 = a_4'$, as a function of the time or of the redshift; $H_4^{(0)}$ its value today, the Hubble constant. Unit: GeV (1/time). Value: $H_4^{(0)} = 100\,h_{100}$ km/s/Mpc. Status: STANDARD.
- σ_{ann}: The annihilation cross-section of the particles χ into Standard-Model particles (through the portals) in the normalisation of FE-61, where n_χ counts the four particle and the four antiparticle states of the bound level: $\langle\sigma_{\mathrm{ann}}\mathsf{v}\rangle$ is a quarter of the annihilation rate $(\sigma\mathsf{v})_{\chi\bar\chi}$ of one particle χ_a with one antiparticle $\bar\chi_a$. Unit: GeV^{-2}. Status: DERIVED.
- v: The relative velocity of two annihilating particles; $\langle\sigma_{\mathrm{ann}}\mathsf{v}\rangle$ is the thermal average of the product. Unit: dimensionless. Status: STANDARD.
- $\langle\cdots\rangle$: The mean (expectation) value of an operator in the state of the medium (the vacuum, a plasma, the Kohn-Sham state). Status: STANDARD.
- $n_{\chi,\mathrm{eq}}$: The number density of χ in thermal equilibrium with the plasma (non-relativistic). Unit: GeV^3. Status: STANDARD.
- g_χ: The number of internal states of the bound level of χ that are in thermal equilibrium. Unit: dimensionless. Status: DERIVED.
- ω_b: The energy of a state of dirac16complex bound to the fold at zero momentum: the four-dimensional mass of the particle χ. Unit: GeV. Status: DERIVED.
- T: The temperature of the Standard-Model plasma (of the photons after the annihilation of electrons and positrons). Unit: GeV. Status: STANDARD.
- π: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.
- e: Euler's number, the base of the natural logarithm; e^X is the exponential of X. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- Ω_χ: The density parameter of the dark matter today. Unit: dimensionless. Value: $\Omega_\chi h_{100}^2 = 0.120$ (Planck 2018). Status: DERIVED.
- h_{100}: The present expansion rate of the fold in units of 100 km/s/Mpc. Unit: dimensionless. Value: 0.674 (Planck 2018). Status: STANDARD.
- x_f: The ratio of the mass of χ to the temperature at which its annihilations stop (freeze-out). Unit: dimensionless. Value: 23 to 30 (for $\omega_\mathrm{b} = 10$ GeV to 10 TeV). Status: STANDARD.
- g_*: The effective number of relativistic degrees of freedom at the freeze-out (g_ρ there). Unit: dimensionless. Status: STANDARD.
- M_Pl: The Planck mass, $M_\mathrm{Pl} = G_\mathrm{N}^{-1/2}$. Unit: GeV. Value: 1.221×10^{19} GeV. Status: STANDARD.

The couplings of χ to the Standard Model (FE-47) and its relic density (FE-61). Status: DERIVED, STANDARD.

requirement	consequence for UM-3	status
the measured density $\Omega_\chi h^2_{100} = 0.120$ by thermal freeze-out	$\langle\sigma_{\rm ann}\mathsf{v}\rangle = 2.2\times10^{-26}$ cm^3/s for $\omega_{\rm b} = 10$ GeV to 10 TeV (FE-61 with $g_\chi = 8$, numerical solution); FE-61 counts the four particle and the four antiparticle states of the bound level in n_χ, so this is a quarter of the annihilation rate of one χ_a with one $\bar\chi_a$, $(\sigma\mathsf{v})_{\chi\bar\chi} = 9\times10^{-26}$ cm^3/s	DERIVED
direct detection, quark portals	a relic through quark portals scatters on nuclei 25 (10 TeV) to 2×10^7 (100 GeV) times above the limit of the LUX-ZEPLIN experiment (4.2 tonne-years of xenon; below its minimum at 40 GeV the limit is not compared): the quark portals are too weak to set the density	ESTIMATE
direct detection, lepton portals	the scattering through a lepton loop and the photon is below the LUX-ZEPLIN limit above about 1.5 TeV	ESTIMATE
direct detection, Higgs portal	the Yukawa coupling $\mathfrak{g}_{h\chi} = -2q_{\rm b}^2/(\omega_{\rm b}v)$ of FE-47 is fixed by the binding: $q_{\rm b}/\omega_{\rm b} \lesssim 0.06$ (100 GeV) to 0.02 (10 TeV)	ESTIMATE
a thin layer and a near-luminal χ	$q_{\rm b} \gg H$ (A-5) and $c_{\rm b}^2 = q_{\rm b}/(q_{\rm b}-H)$ close to 1 ($c_{\rm b}$ the limiting speed of χ along ordinary space in units of the speed of light, the average of the warp $\mathrm{e}^{2H\lvert\sigma\rvert}$ over its profile)	DERIVED (A-5), ESTIMATE ($c_{\rm b}^2$, to first order in the momentum)
the effective number of neutrinos $N_{\rm eff}$ (Planck 2018; ACT DR6, the sixth data release of the Atacama Cosmology Telescope)	the right-handed neutrinos must not be thermalised by the portal: c_X^ν much smaller than the lepton-doublet couplings	DERIVED

$$H \ll q_{\rm b} \lesssim 0.03\,\omega_{\rm b}, \qquad \omega_{\rm b} \gtrsim 1.5\ \text{TeV}, \qquad \mathfrak{g}_{\rm l} \approx \frac{1}{\omega_{\rm b}}\sqrt{\frac{2\pi\cdot 4\langle\sigma_{\rm ann}\mathsf{v}\rangle}{9}}, \qquad c_X^\nu \ll c_X^\ell \tag{A-6}$$

Symbols in (A-6).

- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings and the cosmological constant $\Lambda = -10H^2$; in UM-3 a free parameter. Unit: GeV (1/length). Status: AUTHOR.
- $q_{\rm b}$: The inverse width of a state of dirac16complex bound to the fold: its profile falls as $\mathrm{e}^{-q_{\rm b}\lvert\sigma\rvert}$. Unit: GeV. Status: DERIVED.
- $\omega_{\rm b}$: The energy of a state of dirac16complex bound to the fold at zero momentum: the four-dimensional mass of the particle χ. Unit: GeV. Status: DERIVED.
- $\mathfrak{g}_{\rm q}$, $\mathfrak{g}_{\rm l}$, $\mathfrak{g}_{\rm p}$: The current-portal couplings $\mathfrak{g}_f$ of FE-47 taken equal for all chiral quark states ($\mathfrak{g}_{\rm q}$) or for the nine chiral lepton states that take part, the six of the lepton doublets and the three right-handed charged leptons ($\mathfrak{g}_{\rm l}$), and the coupling to the proton that the lepton loop induces ($\mathfrak{g}_{\rm p}$). Unit: GeV^{-2}. Status: ESTIMATE.
- π: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.
- $\langle\cdots\rangle$: The mean (expectation) value of an operator in the state of the medium (the vacuum, a plasma, the Kohn-Sham state). Status: STANDARD.
- $\sigma_{\rm ann}$: The annihilation cross-section of the particles χ into Standard-Model particles (through the portals) in the normalisation of FE-61, where n_χ counts the four particle and the four antiparticle states of the bound

level: $\langle\sigma_{\mathrm{ann}}\mathsf{v}\rangle$ is a quarter of the annihilation rate $(\sigma\mathsf{v})_{\chi\bar{\chi}}$ of one particle χ_a with one antiparticle $\bar{\chi}_a$. Unit: GeV^{-2}. Status: DERIVED.

- v: The relative velocity of two annihilating particles; $\langle\sigma_{\mathrm{ann}}\mathsf{v}\rangle$ is the thermal average of the product. Unit: dimensionless. Status: STANDARD.
- c_X^f: The current-portal coupling of the half X of dirac16complex with the fermion-number current of the multiplets of kind f (flavour blind); c_X^ℓ and c_X^ν are those of the lepton doublets and of the right-handed neutrinos. Unit: GeV^{-7}. Status: INVENTED.
- X: The label of a half of dirac16complex: $X \in \{\mathrm{L}, \mathrm{R}\}$ (left-handed or right-handed half); a sum over X runs over both. Status: STANDARD.

The window in which dirac16complex is the dark matter of UM-3: a weakly bound state (q_{b} a few per cent of its mass, with the bulk scale H smaller still), heavier than about 1.5 TeV, annihilating through current portals to the nine chiral lepton states that take part (the six of the lepton doublets and the three right-handed charged leptons) with the coupling $\mathfrak{g}_{\mathrm{l}}$ (7.4×10^{-8} GeV^{-2} at 1 TeV, 7.4×10^{-9} GeV^{-2} at 10 TeV), with negligible quark portals and a negligible portal of the right-handed neutrinos. Status: ESTIMATE. Within it the Kohn-Sham states are those of the weakly bound point of the equation of state above: a cold, pressureless gas. Tests: direct detection through the lepton loop at the level 10^{-47} cm^2 for a few TeV (within the reach of the next xenon experiments), and annihilation into charged leptons and neutrinos today, at the rate $(\sigma\mathsf{v})_{\chi\bar{\chi}} \approx 9 \times 10^{-26}$ cm^3/s of one χ_a with one $\bar{\chi}_a$ (four times $\langle\sigma_{\mathrm{ann}}\mathsf{v}\rangle$ of FE-61), to be compared with the limits from gamma rays, positrons and neutrinos from the Galactic centre and dwarf galaxies for a particle that is not its own antiparticle.

3.5 The answer

dirac16complex can be the dark matter of UM-3: its bound states cannot decay into the Standard Model alone, gravitate as matter of the fold, are cold, and have the measured density as a thermal relic, provided they are weakly bound, heavier than about 1.5 TeV and leptophilic, and the right-handed neutrinos stay out of equilibrium. Status: DERIVED (no decay into the Standard Model alone, gravitation, equation of state, relic density), ESTIMATE (the window), OPEN (the stability against the negative-energy states of the indefinite charge).

4 dirac16complex and the dark energy

$$\frac{1}{\kappa_4}\hat{G}^{\mu}{}_{\nu} = \mathcal{T}_{\mathrm{SM}}{}^{\mu}{}_{\nu} + \mathcal{T}_{\mathrm{P}}{}^{\mu}{}_{\nu} + \mathcal{T}_{\mathrm{ff}}{}^{\mu}{}_{\nu} - \frac{V_*}{\kappa}\mathcal{J}^{\mu}{}_{\nu}, \qquad \mathcal{J}^{a}{}_{b} = \kappa\,\mathcal{S}^{a}{}_{b} \text{ (the junction condition)},$$

$$H_4^2 = \frac{8\pi G_{\mathrm{N}}}{3}\big(\rho_{\mathrm{SM}} + \rho_\chi + \rho_\Lambda\big), \qquad 8\pi G_{\mathrm{N}} = \frac{\kappa_4}{1 + 5\epsilon_{\mathrm{r}}/6}, \qquad \epsilon_{\mathrm{r}} = \frac{\kappa_4 V_*}{\kappa H}, \qquad \rho_\Lambda = \rho_{\mathrm{ff}} - \frac{31 H V_*}{4\kappa}, \tag{FE-59}$$

$$\frac{\hat{R}}{2\kappa_4} + q_{\mathrm{SM}} + q_\chi + q_{\mathrm{P}} + q_{\mathrm{ff}} = \frac{V_*}{\kappa}\mathcal{J}^{(r)}{}_{(r)}, \qquad q_{\mathrm{SM}} = \big\langle \mathcal{L}_{\mathrm{H}} + \mathcal{L}_{\mathrm{gauge}} + \mathcal{L}_{\mathrm{f}} + \mathcal{L}_{\mathrm{Y}} \big\rangle, \qquad q_{\mathrm{P}} = \langle \mathcal{L}_{\mathrm{P}} \rangle$$

Symbols in (FE-59).

- κ_4: The coupling of the induced gravity of the fold, $\kappa_4 = 8\pi G_{\mathrm{ind}}$; four-dimensional gravity on the fold has $8\pi G_{\mathrm{N}} = \kappa_4/(1 + 5\epsilon_{\mathrm{r}}/6)$ (the Friedmann equation of the fold to first order in ϵ_{r}). Unit: GeV^{-2}. Value: $\kappa_4 \simeq 8\pi G_{\mathrm{N}} = 8\pi/M_{\mathrm{Pl}}^2$. Status: ASSUMED.
- $\hat{G}^{\mu}{}_{\nu}$: The Einstein tensor of the four-dimensional metric $\hat{h}_{\mu\nu}$ of the fold. Unit: $1/\mathrm{length}^2$. Status: STANDARD.
- μ: A four-dimensional coordinate index running over the observable coordinates x_1, x_2, x_3, x_4 of the fold (signature $+,+,+,-$); a repeated index is summed. Status: STANDARD.
- ν: A four-dimensional coordinate index (as μ). Status: STANDARD.
- $\mathcal{T}_{\mathrm{SM}}$: The energy-momentum tensor of the Standard Model on the fold, $\mathcal{T}_{\mathrm{H}} + \mathcal{T}_{\mathrm{gauge}} + \mathcal{T}_{\mathrm{f}}$. Unit: GeV^4. Status: STANDARD.
- $\mathcal{T}_{\mathrm{P}}$: The energy-momentum tensor of the portals of dirac16complex. Unit: GeV^4. Status: DERIVED.
- $\mathcal{T}_{\mathrm{ff}}{}^{a}{}_{b}$: The energy-momentum tensor of the fold fluid (frame components of the fold, four-dimensional normalisation). Unit: GeV^4. Status: ASSUMED.
- V_*: The physical volume of the extra times x_5, x_6, x_7 on the fold in the late universe, $V_* = V_{\mathrm{t}}(\mathrm{e}^{-a_4}b_4)^3$ at a time when $\mathrm{e}^{-a_4}b_4$ is constant; it converts the eight-dimensional normalisation into the four-dimensional one. Unit: length^3 (GeV^{-3}). Status: ASSUMED.
- κ: The eight-dimensional gravitational coupling, $\kappa = 8\pi G_8$. Unit: GeV^{-6}. Status: AUTHOR.

- $\mathcal{J}^a{}_b$: The jump of the junction potential across the fold computed from the metric: the geometric side of the junction condition $\mathcal{J} = \kappa\mathcal{S}$, which equates it to κ times the energy-momentum tensor $\mathcal{S}$ of everything on the fold. Unit: GeV. Status: DERIVED.
- a: A frame (orthonormal) index running over $0, 1, \ldots, 7$ (frame direction (0) along σ, (1), (2), (3) along ordinary space, (4) along the time, (5), (6), (7) along the extra times); raised and lowered with η_{ab}; on the fold ($a, b \in \{1, \ldots, 7\}$) the frame of the fold; a repeated index is summed. Status: STANDARD.
- b: A frame index (as a). Status: STANDARD.
- $\mathcal{S}^a{}_b$: The thin source of the fold: its energy-momentum tensor per unit proper length across the fold (eight-dimensional normalisation, frame components along the fold); $\kappa\mathcal{S}$ is the jump of the potential $\mathfrak{F}$. Unit: GeV^7. Status: DERIVED.
- H_4: The expansion rate (Hubble rate) of the fold, $H_4 = a_4'$, as a function of the time or of the redshift; $H_4^{(0)}$ its value today, the Hubble constant. Unit: GeV (1/time). Value: $H_4^{(0)} = 100\,h_{100}$ km/s/Mpc. Status: STANDARD.
- π: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.
- G_N: Newton's constant of four-dimensional gravity on the fold, $G_\mathrm{N} = 1/M_\mathrm{Pl}^2$. Unit: GeV^{-2}. Value: $6.708{\times}10^{-39}$ GeV^{-2}. Status: STANDARD.
- ρ_SM: The energy density of the Standard-Model particles. Unit: GeV^4. Status: STANDARD.
- ρ_χ: The energy density of the dark matter, the bound states χ of dirac16complex; $\rho_\chi^{(0)}$ its value today. Unit: GeV^4. Status: DERIVED.
- ρ_Λ: The vacuum energy density of the fold, the dark energy of UM-3: $\rho_\Lambda = \rho_\mathrm{ff} - 31HV_*/(4\kappa)$ (FE-59). Unit: GeV^4. Status: DERIVED.
- ϵ_r: The bookkeeping parameter of the response of the bulk to the matter of the fold: the ratio of the bulk to the fold gravity, $\epsilon_\mathrm{r} = \kappa_4 V_*/(\kappa H)$. Unit: dimensionless. Value: $\lesssim 10^{-15}$ (an estimate: if a fraction ϵ_r of the gravitational waves of the fold travelled with the bulk gravitons, whose speed squared is 15/2 times that of light, their speed would differ from that of light by about $3\epsilon_\mathrm{r}$, and the gravitational waves and the light of the neutron-star merger GW170817 arrived together to about 10^{-15}). Status: DERIVED (the definition), ESTIMATE (the bound).
- H: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of x_0, the Lovelock couplings and the cosmological constant $\Lambda = -10H^2$; in UM-3 a free parameter. Unit: GeV (1/length). Status: AUTHOR.
- ρ_ff: The energy density of the fold fluid. Unit: GeV^4. Value: $31HV_*/(4\kappa)$ at the static block. Status: ASSUMED.
- $\hat{R}$: The scalar curvature of the metric $\hat{h}_{\mu\nu}$ of the fold. Unit: 1/length^2. Status: STANDARD.
- q_SM: The pressure of the Standard Model along the extra times at the fold: the mean on-shell Lagrangian density of the Standard-Model fields (the extra-time component of the energy-momentum tensor of every term of the fold is its Lagrangian, because the fields of the fold do not depend on the extra times). Unit: GeV^4. Status: DERIVED.
- q_χ: The pressure of the bound states of dirac16complex along the extra times, integrated across the fold (their on-shell Lagrangian, the self-interaction energy per volume; zero for a dilute gas). Unit: GeV^4. Status: DERIVED.
- q_P: The pressure of the portals along the extra times: their mean on-shell Lagrangian density. Unit: GeV^4. Status: DERIVED.
- q_ff: The pressure of the fold fluid along the extra times. Unit: GeV^4. Value: $-61HV_*/(8\kappa)$ at the static block. Status: ASSUMED.
- r: An extra-time direction of the fold: $r \in \{5, 6, 7\}$ (x_5, x_6, x_7); not summed (the three components are equal). Status: STANDARD.
- $\langle\cdots\rangle$: The mean (expectation) value of an operator in the state of the medium (the vacuum, a plasma, the Kohn-Sham state). Status: STANDARD.
- $\mathcal{L}_\mathrm{H}$: The Lagrangian of the Higgs doublet on the fold. Unit: GeV^4. Status: AUTHOR.
- $\mathcal{L}_\mathrm{gauge}$: The Lagrangian of the gauge fields of $SU(3)_C \times SU(2)_L \times U(1)_Y$ on the fold. Unit: GeV^4. Status: STANDARD.
- $\mathcal{L}_\mathrm{f}$: The kinetic Lagrangian of the quarks and leptons on the fold. Unit: GeV^4. Status: STANDARD.
- $\mathcal{L}_\mathrm{Y}$: The Yukawa Lagrangian of the quarks and leptons with the Higgs doublet. Unit: GeV^4. Status: STANDARD.
- $\mathcal{L}_\mathrm{P}$: The Lagrangian of the portals of dirac16complex at the fold. Unit: GeV^4. Status: INVENTED.

The Einstein equations of the fold (FE-59): the dark energy of UM-3 is the vacuum energy of the fold $\rho_\Lambda =$

$\rho_{\mathrm{ff}} - 31HV_*/(4\kappa)$, the energy of the fold fluid minus the part that the kink of the static block needs. Status: DERIVED (with the ASSUMED fold fluid).

$$\rho_\Lambda = \Omega_\Lambda\, \rho_{\mathrm{c}}^{(0)} = 0.685 \cdot \frac{3\big(H_4^{(0)}\big)^2}{8\pi G_{\mathrm{N}}} = 2.5 \times 10^{-47}\ \mathrm{GeV}^4 = \big(2.2\ \mathrm{meV}\big)^4 \tag{A-7}$$

Symbols in (A-7).

- ρ_Λ: The vacuum energy density of the fold, the dark energy of UM-3: $\rho_\Lambda = \rho_{\mathrm{ff}} - 31HV_*/(4\kappa)$ (FE-59). Unit: GeV^4. Status: DERIVED.
- Ω_Λ: The density parameter of the dark energy (the vacuum energy of the fold) today. Unit: dimensionless. Value: 0.685 (Planck 2018). Status: DERIVED.
- ρ_{c}: The critical density, the energy density of a spatially flat fold at the expansion rate H_4; $\rho_{\mathrm{c}}^{(0)}$ its value today. Unit: GeV^4. Value: $\rho_{\mathrm{c}}^{(0)} = 1.0537 \times 10^{-5}\, h_{100}^2$ GeV/cm^3. Status: STANDARD.
- H_4: The expansion rate (Hubble rate) of the fold, $H_4 = a_4'$, as a function of the time or of the redshift; $H_4^{(0)}$ its value today, the Hubble constant. Unit: GeV (1/time). Value: $H_4^{(0)} = 100\, h_{100}$ km/s/Mpc. Status: STANDARD.
- π: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.
- G_{N}: Newton's constant of four-dimensional gravity on the fold, $G_{\mathrm{N}} = 1/M_{\mathrm{Pl}}^2$. Unit: GeV^{-2}. Value: $6.708{\times}10^{-39}$ GeV^{-2}. Status: STANDARD.

The vacuum energy that the measurements require ($H_4^{(0)} = 67.36$ km/s/Mpc $= 1.44 \times 10^{-42}$ GeV). Status: STANDARD.

Can dirac16complex be the dark energy? A dark energy needs a pressure close to minus its energy density, an equation of state $w = P/\rho$ (pressure over energy density) close to -1. The gas of bound states has the pressure $P_{\mathrm{KS}} = n_\chi \mathsf{E}_{\mathrm{int}}$ of its self-interaction (A-8), which is positive for a repulsive self-interaction and vanishes in the dilute limit ($w_\chi = 7.2 \times 10^{-8}$ to 4.7×10^{-5} at $G_{\mathrm{KS}} = 10^{-6}$, A-8). The gas of χ is dark matter, not dark energy. The other contribution of dirac16complex to the vacuum energy, the energy of its unoccupied and its negative-energy states (the sea, which the Kohn-Sham states leave out), is a renormalisation of ρ_Λ that UM-3 does not compute; its sign and size are OPEN, and with the indefinite Krein norm it is not even clear that it is bounded. The dark energy of UM-3 is therefore the vacuum energy of the fold, a free parameter: its value $(2.2\ \mathrm{meV})^4$ is accommodated by choosing ρ_{ff}, not explained, and it has $w = -1$ to the order of the power series computed here (first order in $\epsilon_{\mathrm{r}} \lesssim 10^{-15}$). With the Planck 2018 parameters UM-3 reproduces the CMB, but it keeps the tension of the cosmological-constant model with the BAO of DESI DR2 (2.3 standard deviations in DESI's analysis) and cannot produce the evolving dark energy that DESI DR2 with the CMB prefers at 3.1 standard deviations: an equation of state $w(a) = w_0 + w_a(1-a)$ that changes with the scale factor $a = \mathrm{e}^{a_4}$ of the fold, with $w_0 > -1$ and $w_a < 0$. Status: DERIVED (`um3_kohnsham.py`, K5, K7; `um3_cosmology.py`, C5: $w = -1$), STANDARD (DESI's significances), OPEN (the sea).

4.1 The answer

No: dirac16complex is not the dark energy of UM-3. Its bound states are cold dark matter; the dark energy is the vacuum energy of the fold, a free parameter with $w = -1$.

5 What the author may have left out

The author asked to be told if something important was missing from his specification. UM-3 had to add the following, and the analysis found the following problems.

item	why it is needed or what it means	where it stands
right-handed quark singlets and the Yukawa couplings	the quark doublets alone have no masses and the hypercharges cancel the gauge anomalies only with the singlets	added (a default of UM-3, proposed in the notification of 17:20, which the author did not answer: ASSUMED)

item	why it is needed or what it means	where it stands
three right-handed neutrinos	Dirac neutrino masses and the mixing that flavour conversion needs; the anomalies still cancel	added (accepted by the author, 17:32); they must never thermalise
the place of the Standard Model	in the bulk, every species would move with its own limiting speed and three neutrino profiles would oscillate wrongly; on the fold all share one light cone	the Standard Model lives on the fold (ASSUMED, motivated by the DERIVED limiting speeds)
a thin source at the fold	the kink of the warp $\mathrm{e}^{-H\lvert\sigma\rvert}$ requires one (the fold fluid); its energy above the static value is the dark energy	ASSUMED
an Einstein-Hilbert term of the fold	bulk gravitons move 2.74 times faster than light at the static block and have no time-kinetic term at the author's vacuum (the history $a_4 = Hx_4$ with constant b_4, which needs no sources in the bulk); gravitational waves move with the speed of light to 10^{-15}	ASSUMED (induced gravity)
a medium of the bulk	the static late-time bulk and the frozen extra times need bulk matter that violates the null energy condition	ASSUMED, no field theory given
the response of that medium	to first order in the bulk response, the component of the junction condition along the extra times (the radion condition) cannot be met by a rigid medium without changing the expansion law of the fold (dust would expand as $x_4^{5/6}$); a responding medium meets it with the radion mode, a combination of first-order jumps of the metric at the fold in the ratio $1 : -1 : 5$ that changes only that component	ASSUMED (the radion mode)
frozen extra times in the late universe (the size $\mathrm{e}^{-a_4}b_4$ of the extra times constant)	otherwise the couplings of the Standard Model change: in the author's vacuum ($a_4 = Hx_4$, b_4 constant) the gauge and Yukawa couplings grow as $\mathrm{e}^{3Hx_4/2}$, the quartic Higgs coupling and Newton's constant as e^{3Hx_4}	required
a gauge field for the ad hoc Higgs vacuum	the literal ad hoc vacuum carries an isospin charge that the Gauss law forbids; with the pure-gauge $W_{x_4}^2$ it is a rotated vacuum with one radial residual, and it must decay before nucleosynthesis	DERIVED
the Krein norm of dirac16complex	the signature $(4, 4)$ of the author's matrices makes the charge indefinite: half of the bound states carry negative energy	OPEN (stability)

item	why it is needed or what it means	where it stands
the extra times	fields that depend on them are tachyonic	ASSUMED (no dependence), OPEN
Lorentz violation in the bulk	the particles χ move faster than light on the fold unless $q_{\mathrm{b}} \gg H$	a condition on the parameters
direct detection	the Higgs portal fixed by the binding and the quark portals are strongly constrained	a weakly bound, leptophilic χ above about 1.5 TeV
the dark energy	a vacuum energy with $w = -1$; DESI DR2 prefers an evolving one	OPEN (tension at 2.3 to 3.1 standard deviations)
the primordial era	the transition from the author's vacuum to the late universe and the primordial spectrum are not computed	OPEN (Planck inputs)
the lightest neutrino mass and the Dirac phase	not fixed by the specification	ASSUMED: $m_1 = 0$, phase 0